Examples and non-examples

Clarifying what counts

Use examples and non-examples to make the defining feature of a concept clear. Carefully chosen near-misses help students stop relying on superficial appearance.

How to do it well

1 Identify the critical feature. Name the single feature that determines whether something belongs to the concept.

2 Choose a near-miss. Use a non-example that shares the example's surface features but removes the critical feature. Avoid random or obviously wrong cases.

3 Alternate and vary. Show an example, then a related non-example; change one feature at a time so the contrast is visible.

4 Include boundary cases. Use cases that are just inside or just outside the concept so students must apply the definition, not guess from appearance.

5 Pause, commit, reveal. Ask what changed and what stayed the same; have every student decide before revealing and explaining the answer.

What this sounds like

"Both diagrams look similar. Which is a circuit, and which feature decides? Explain the smallest change that would make the other one a valid example."

Classroom example. A Maths teacher contrasts a valid rectangle with a near-miss that shares several visual features. Pupils identify the defining properties rather than rely on familiar orientation. A tilted valid example and a plausible invalid case follow. Students justify each classification, making the concept boundary clearer than a page of identical conventional examples.

Top tips that make the technique work

Choose close contrasts. A near-miss should share enough features with the example to make the boundary worth thinking about.

Change one important feature. Too many simultaneous changes make it difficult to see what determines membership.

Include unusual valid cases. Prevent students from treating the first familiar example as the definition.

Ask for the criterion. "It looks right" is less revealing than a reason linked to the critical feature.

Return to the definition. After comparisons, students should be able to describe what counts and test a new case against it.

Do and avoid

Do	Avoid
Use boundary cases and independent decisions.	Obviously irrelevant wrong examples.
Connect the judgement to the defining feature.	Changing several features so the critical contrast disappears.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Non-Examples